

Jihwan Kim

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EDUCATION

Ph.D. in Physical Oceanography

2019–

School of Earth and Environmental Sciences, College of Natural Sciences, Seoul National Univ.

2023

Graduated as the youngest and fastest to complete a Ph.D. in Physical Oceanography in Korea

B.S. in Earth Science Education

2013–

College of Education, Seoul National University

2018

**Compulsory Military Service in 2015–2016

RESEARCH INTERESTS

- Pelagic fisheries and upper-ocean current systems in the Northwestern Pacific
- Statistical forecasting methods for fisheries and environmental data
- Machine learning applications for sustainable fishing practices, including super-resolution of coarse spatial-temporal data and forecasting future fishing grounds

SKILLS

Languages:

Native Korean; Professional Working Proficiency in English and Japanese.

Technical Proficiencies:

- **Core Programming & Scientific Computing:**
 - **Advanced Python:** for complex data manipulation, modeling, and analysis.
 - **Advanced MATLAB:** for oceanographic data processing and algorithm development.
 - **Proficient in Julia & Fortran:** Experience in leveraging for high-performance scientific computations.
- **Data Analysis, Modeling & Visualization:**
 - **Spatio-temporal Data Analysis:** Specialized in techniques like EOF and CSEOF for oceanographic and fisheries data.
 - **Statistical Forecasting & Regression Analysis:** Applying predictive models to environmental and fisheries datasets.
 - **Machine Learning:** Development of time-series prediction models using TensorFlow and PyTorch for applications like fishing ground forecasting and data super-resolution.
 - **Geospatial Information Systems (GIS):** Proficient in GIS software for mapping and spatial analysis of catch/effort data.

- **Data Visualization:** Creating informative visualizations from complex datasets
- **Data Management & Web Technologies:**
 - **Database Management:** Proficient with relational databases (MySQL) for data storage, querying, and warehousing.
 - **Dashboard Development:** Experience in creating interactive data dashboards
 - **Front-End Familiarity:** Exposure to front-end frameworks (e.g., Vue, React) for specific project components or data presentation layers.
- **General Applications:**
 - Proficient with MS Office Suite (Word, Excel, PowerPoint) for reporting and presentations.

TEACHING EXPERIENCE

● **Teaching Assistant at Seoul National University**

Physical Oceanography (Spring 2019)

Observation and Comprehension of Natural Disasters (Fall 2019, Spring 2020)

Theory and Practice in Computational Sciences 1 (Spring 2022)

● **Lectures at Seoul National University**

Computational Science: Subject Research (The Missing Semester of CS Education) (Spring 2021)

HONORS AND AWARDS

● **JOISS Marine Science Big Data Competition**

2021–2022

2nd Prize rewarded by Chairman of the Korea Oceanographic Society (2021 and 2022, respectively). JOISS Marine Science Big Data Competition is selected as the UN Ocean Decade Activity.

● **Succeeding Generations of Fundamental Sciences Fellowship**

2019–2020

Selected in the field of Oceanography. Seoul National University awarded a full scholarship to graduate students who can be a succeeding leader of education and research in fundamental sciences.

● **Travel Grants for Young Scientists**

PICES 30th annual meeting in Busan, Republic of Korea (2022)

EXPERIENCE

- **Ocean Engineer, Sustainable Fisheries Management Team, Collecte Localisation Satellites (CLS) – France & Japan** 2025–
 - Conduct oceanographic analyses to optimize tuna fishing vessel operations, ensuring compliance with safety regulations and sustainable fishing guidelines.
 - Develop and apply predictive models using real-time oceanographic data (e.g., SST, currents, chlorophyll) to monitor tuna fishing activities, with the objective of enhancing operational efficiency and fuel conservation for client vessels.
 - Contribute to the ongoing design and implementation of the CATSAT system, a satellite-based fisheries monitoring solution. Specific responsibilities include algorithm development for oceanographic data integration and defining data requirements for predictive features, supporting improved vessel tracking and fisheries management decision-making.

- **Data & Fisheries Science Advisor to the North Pacific Fisheries Commission** 2024–
 - **Resource Management - Pacific Saury:**
 - Delivered presentations and prepared publications on the spatial-temporal variability in catch per unit effort of Pacific saury (*Cololabis saira*) and its relationship with interannual basin-scale environmental variability in the North Pacific.
 - Presented research on stochastic statistical simulations for forecasting the spatial-temporal distribution of Pacific saury based on observational data in the Northwestern Pacific.
 - Developed and presented the "Physically Consistent Super-Resolution of Pacific Saury Spatio-Temporal Distribution Using Multi-head Attention U-Net."
 - **Resource Management - Chub Mackerel:**
 - Investigated and reported on the effects of Kuroshio Current variability and the Pacific Decadal Oscillation on the recent decline in Chub Mackerel (*Scomber japonicus*) catch in the Northwestern Pacific during the 2020s.
 - **GIS & Data Integration:**
 - Developed a GIS-based mapping system to visualize and analyze Neon Flying Squid catch and effort data submitted by North Pacific Fisheries Commission members.
 - **Compliance Analytics & Reporting:**

- Designed matching algorithms to integrate Vessel Monitoring System (VMS) data with submitted transshipment documents, high seas boarding inspection reports, and Automatic Identification System data, thereby detecting potential non-compliance.
 - Produced and presented comprehensive reports on Fisheries Overview, Transshipment Overview, and VMS Overview at North Pacific Fisheries Commission Meetings in Osaka (2024 and 2025).
 - Compiled the 2024 North Pacific Fisheries Commission Fisheries and Marine Resources Inventories for inclusion in the Food and Agriculture Organization's Fisheries and Resources Monitoring System.
- **Postdoctoral Fellowship in the North Pacific Fisheries Commission and Tokyo University of Marine Science and Technology** **2023–2024**
 - **Resource Management - Pacific Saury:**
 - Prepared a presentation and publication titled "The Interannual to Decadal Relationship Between Total Catch Variability of Pacific Saury (*Cololabis saira*) and Basin-Scale Ocean Environmental Variability in the North Pacific."
 - **Resource Management - Chub Mackerel:**
 - Prepared a presentation and publication on "Assessment of the Interannual to Decadal Relationship Between Catch and Population Variability of Chub Mackerel (*Scomber japonicus*) in Japan and Basin-Scale Ocean Environmental Variability in the North Pacific."
 - **Catch Reporting Measures:**
 - Developed strategic measures for reprinting catches to improve accuracy in fisheries reporting.
 - **GIS-Based Mapping:**
 - Designed and implemented a GIS-based mapping system for Neon Flying Squid catch and effort data from North Pacific Fisheries Commission members.
 - **IUU Fishing Measures:**
 - Developed algorithms for the Vessel Monitoring System to detect IUU fishing vessels.
 - Created algorithms to integrate and match air surveillance data with VMS data for enhanced detection and monitoring.
- **High school Science Teacher** **2018–2019**

Earth Science Teacher at Sehwa High School, Seoul, Republic of Korea

- **Auxiliary Policeman (Compulsory Military Service)** 2015–2016
Strike Force of the Police in Gangwon Province, Republic of Korea

EXPERIENCE IN FUNDED PROJECT

- **A study on upstream-downstream connectivity of the western boundary current in the North Pacific and its relationship with mesoscale eddy activity** 2019–2023
funded by the National Research Foundation of the Republic of Korea
investigates the Kuroshio current variability and its ecological impact variability
- **Seychelles-Chagos Thermocline Ridge Pilot Time-series Observation Initiative**
funded by the Korea Institute of Ocean Science Technology
investigates upwelling and related ecological variability in Seychelles-Chagos Thermocline Ridge
- **East Asian Seas Time-series** 2021– 2023
funded by the Ministry of Land, Transport and Maritime Affairs, Korea
investigates ocean-environmental variability (physical, biological, etc.) in the East/Japan Sea

FIELD EXPERIENCE

- **The East China Sea and Yellow Sea (CTD, ADCP, DO sampling)**
R/V ONNURI (Jul 2022)
- **The Korea/Tsushima Strait (Induced voltage measurement on submarine cable)**
Submarine cable voltage measurement between Busan, Republic of Korea and Hamada, Japan. Volume transport through the Korea/Tsushima Strait can be estimated from the motion-induced voltage on submarine cable across the strait. Submarine cable voltage observations were stopped on May 17, 2017 owing to a measurement equipment failure and re-initiated on Dec 13, 2020. Jihwan Kim devised and built a new observation solution, a new data analysis method, and a cloud data server.

PRESENTATIONS

- **2024 Vessel Monitoring System Overview** Mar, 2025
8th TCC Meeting, Osaka, Japan, Mar 20, 2025
- **2024 Transshipment Overview**
8th TCC Meeting, Osaka, Japan, Mar 19, 2025
- **Revised GIS-based mapping system for Neon Flying Squid (*Ommastrephes bartramii*) catch and effort data from members of the North Pacific Fisheries Commission** Dec, 2024
14th SSC PS meeting, Tokyo, Japan, Dec 18, 2024

- **Spatial-temporal variability in catch per unit effort of Pacific saury (*Cololabis saira*) and its relationship to interannual basin-scale environmental variability in the North Pacific** Aug, 2024
13th SSC PS meeting, Online, Aug 27, 2024
- **Effects of Kuroshio Current Variability and Pacific Decadal Oscillation on Recent Decline in Chub Mackerel (*Scomber japonicus*) Catch in the Northwestern Pacific in the 2020s** Jul, 2024
9th TWG CSMA Meeting, Yokohama, Japan, Jul 19, 2024
- **Vessel Monitoring System Overview from 2022 to 2023** Apr, 2024
7th TCC Meeting, Osaka, Japan, Apr 9, 2024
- **A GIS-based mapping system for Neon Flying Squid (*Ommastrephes bartramii*) catch and effort data from members of the North Pacific Fisheries Commission** Dec, 2023
12th SSC PS meeting, Nanaimo, Canada, Dec 18, 2023
- **The interannual to decadal relationship between total catch variability of Pacific saury (*Cololabis saira*) and basin-scale ocean environmental variability in the North Pacific** Dec, 2023
12th SSC PS meeting, Nanaimo, Canada, Dec 12, 2023
- **Monitoring volume transport through measurement of cable voltage across the Korea Strait Since November 2020** Nov, 2022
Korea Oceanographic Society Autumn Conference, Gangneung, Republic of Korea, Nov 2, 2022
- **Interannual variability of barotropic sea level difference across the Korea/Tsushima Strait and its relationship to upper-ocean currents variability in the western North Pacific** Sep, 2022
PICES-2022 annual meeting, Busan, Republic of Korea, Sep 30, 2022
- **Interannual variability of barotropic sea level difference across the Korea/Tsushima Strait and its relationship to upstream Kuroshio variability** Mar, 2022
Ocean Sciences Meeting, Online Everywhere, Mar 3, 2022
- **Interannual variability of Yellowfin tuna (*Thunnus albacares*) and Bigeye tuna (*Thunnus obesus*) catches in the southwestern tropical Indian Ocean and its relationship to climate variability** Dec, 2020
AGU Fall Meeting, Online Everywhere, Dec 7, 2020
- **Potential Predictability of Skipjack Tuna (*Katsuwonus pelamis*) Catch in the Western Central Pacific** Oct, 2019
Korea Oceanographic Society Autumn Conference, Gangneung, Republic of Korea, Oct 31, 2019

PUBLICATIONS

- **Jihwan Kim** and Hanna Na (2022), Interannual Variability of Yellowfin Tuna (*Thunnus albacares*) and Bigeye Tuna (*Thunnus obesus*) Catches in the Southwestern Tropical Indian Ocean and Its Relationship to Climate Variability, *Frontiers in Marine Science*, **9**, 857405
- **Jihwan Kim**, Hanna Na, Young-Gyu Park, and Young Ho Kim (2020), Potential Predictability of Skipjack Tuna (*Katsuwonus pelamis*) Catches in the Western Central Pacific, *Scientific Reports*, **10**, 3193.

MANUSCRIPT SUBMITTED

- **Jihwan Kim**, Libin Dai, Christopher N. Rooper, Yi-Jay Chang, Jhen Hsu, Hyejin Song, Vladimir Kulik, Mei-chin Juan, Robert Day, Aleksandr Zavolokin, Rentaro Mitsuyu, and Toshihide Kitakado, Spatio-temporal variability of density distribution of Pacific saury (*Cololabis saira*) and its relationship to basin-scale Ocean environmental variability in the North Pacific
- **Jihwan Kim**, Christopher N. Rooper, Shota Nishijima, Kazuhiro Oshima, Robert Day, and Aleksandr Zavolokin, Effects of Kuroshio Current Variability and Pacific Decadal Oscillation on Recent Decline in Chub Mackerel (*Scomber japonicus*) Catch in the Northwestern Pacific in the 2020s

MANUSCRIPT IN PREPARATION

- **Jihwan Kim**, Libin Dai, Christopher N. Rooper, Shinichiro Nakayama, Yi-Jay Chang, Jhen Hsu, Hyejin Song, Vladimir Kulik, Mei-chin Juan, Robert Day, Aleksandr Zavolokin, Rentaro Mitsuyu, and Toshihide Kitakado, Stochastic statistical simulations of spatial-temporal distribution of Pacific saury (*Cololabis saira*) and its forecast based on observational data in the Northwestern Pacific
- **Jihwan Kim** and Hanna Na, Assessment of the Favorable Ocean Conditions and Potential Predictability of Annual Catch amounts of Skipjack tuna (*Katsuwonus pelamis*), Yellowfin tuna (*Thunnus albacares*), and Bigeye Tuna (*Thunnus obesus*) in the Western Central Pacific